SciGraph-LLM: Automatic Knowledge Graph Construction from Scientific Papers

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Abstract

The automatic extraction of structured scientific knowledge from literature is hampered by the implicit, variably reported nature of claims and their evidential grounding challenges unaddressed by conventional knowledge graphs, which lack the capacity to represent argumentative structure and provenance. This paper proposes a constrained, evidence-grounded pipeline that transforms unstructured PDFs of scientific papers into provenance-aware scientific knowledge graphs using large language models (LLMs) under strict textual fidelity constraints. By segmenting documents into overlapping windows and applying GPT-5 with schema-enforced prompts, the system extracts atomic claims, their supporting evidence spans, methods, datasets, metrics, and numerical results—all linked to verbatim source text. Post-processing via entity canonicalization and relation extraction yields a heterogeneous knowledge graph where each edge is supported by direct evidence. Evaluated on ten LLM-KG papers, the pipeline achieves 81.3% evidence precision and 59.9% claim F1 with full structural alignment, significantly outperforming unconstrained baselines in reliability. Results demonstrate that strict grounding enables auditable, reproducible knowledge extraction, but also reveals limitations in handling non-contiguous evidence and implicit reasoning. The results highlight the potential of structured claim-evidence representations to support transparent and traceable scientific knowledge organization, while indicating the continuing importance of human oversight in maintaining factual reliability.



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1 Introduction

The rapid growth of scientific literature has created a persistent challenge for researchers, policy-makers, and automated systems that rely on structured knowledge. Scholarly articles contain rich information on hypotheses, experimental designs, results, limitations, and claims supported by evidence. However, this information is typically embedded in complex narrative structures that span multiple sections and rhetorical functions, making it difficult to retrieve and analyze systematically. Traditional knowledge graph (KG) construction approaches, which are often optimized for triple extraction from general text, provide limited support for capturing scientific argumentation, methodological context, and evidence grounding [2, 44]. As a result, extracting reliable, interpretable, and context-aware scientific knowledge from research publications remains a largely open problem.

Large language models (LLMs) have recently achieved strong results in tasks involving text understanding, information extraction, and structured reasoning [23, 30]. Emerging work has explored their use for KG population, domain-specific knowledge organization, and retrieval-augmented generation. These developments suggest opportunities for supporting automated discovery, literature synthesis, and scientific data management. Yet, current approaches are

typically evaluated on short or semi-structured documents, operate primarily at the sentence level, and emphasize entity–relation triples without capturing richer scientific structures such as claims, supporting evidence, and methodological provenance. Moreover, fully automated LLM extraction introduces risks related to factuality, hallucination, and incomplete context modeling, which are especially consequential in scientific domains.

At the same time, recent research has highlighted the value of human–AI collaboration for KG validation, showing that expert involvement can support reliability, especially in domains where errors carry scientific or practical implications [11, 15]. However, most existing human-in-the-loop designs focus on post-hoc validation or annotation rather than collaborative, traceable construction of structured scientific knowledge. This gap limits the deployment of KG-based tools for literature analysis and knowledge discovery in research workflows.

To address these limitations, recent work has begun to consider representations beyond isolated triples [42], including structured scientific claims and associated evidence. Building on these developments, the present study examines an approach for constructing scientific claim-evidence knowledge graphs from research articles using LLM-based extraction, retrieval mechanisms, and selective human verification. Scientific text is modeled not simply as a source of entities and relations, but as a structured collection of claims linked to supporting sentences, experimental settings, and contextual qualifiers. This formulation aims to support transparency and traceability by retaining explicit links to primary textual sources.

2 Related Work

Research at the intersection of large language models (LLMs) and knowledge graphs (KGs) is developing along several complementary directions [17, 39, 45]. The conceptual map in Figure 1 illustrates the current research landscape at the intersection of large language models and knowledge graphs.

One group of studies focuses on using LLMs to extract structured facts from unstructured text and align entities across heterogeneous sources. For instance, Yang et al. [37] combine neural retrieval with LLM-based reasoning to support interpretable entity alignment across KGs, while Haque et al. [9] examine how open-weight language models can generate class descriptions and RDF triples for existing ontologies. Similar observations about the capacity of compact models for structured extraction tasks are reported by Gajo and Barrón-Cedeño [7], who analyze the behavior of smaller LLMs in controlled extraction settings. These works illustrate shared interest in linking symbolic structures and language-based models, yet they operate primarily on general KG resources or curated benchmarks rather than on full scientific texts.

A second line of work considers domain-oriented KG construction. Elliott et al. [5] study biomedical knowledge represented in scientific abstracts, extracting proposition-level statements to form enriched triples. In the context of urban planning, Zhang et al. [40] structure master-plan documents into thousands of triplets to enable spatial reasoning and policy analysis. Shao et al. [27] extend traditional triple formats with richer tuple structures to represent manufacturing knowledge, and Deng et al. [3] transform medical knowledge graphs into QA corpora to align LLMs with domain facts. Work in medical question answering by Wang et al. [34] also employs entity linking and multi-hop reasoning over domain KGs. Although these studies address specialized technical domains, they typically rely on structured or semi-structured documents, whereas scientific articles tend to combine argumentative narrative, methodological descriptions, and empirical evidence in more varied forms.

Another relevant stream investigates ways to incorporate human judgement into KG-centric workflows. Tsaneva et al. [32] compare automated, expert-only, and hybrid validation modes when verifying large-scale scholarly KGs, noting trade-offs in precision and recall across settings. In chemistry, Fu et al. [6] combine structured catalytic knowledge with LLM-based text generation and expert-defined scoring to assist pathway identification, and Marandi et al. [21] employ LLMs to extract system relationships from technical documents and support KG-based diagnostic reasoning. These works point to increasing attention on combining algorithmic extraction with targeted human oversight, particularly where domain knowledge or interpretability are critical.

Some studies investigate procedural, causal, and multimodal knowledge representations. Li et al. [16] construct multimodal causal hypergraphs for equipment fault analysis, Singh and Hsieh [29] apply chained prompting to normalize and generate construction workflows, Shi et al. [28] combine KG retrieval with chain-of-thought prompting for industrial documentation, and Liu et al. [18] build 3D scene graphs from drone imagery to support infrastructure inspection. These works highlight needs that arise when reasoning spans multiple modalities, steps, or contextual elements.

Across these research directions, there is a shared emphasis on connecting unstructured language with structured representations, and on balancing automation with interpretability [4] and domain knowledge [36]. However, comparatively little work addresses the extraction and structuring of scientific knowledge from full research papers, where claims, evidence, experimental context, and methodological details must be identified, linked, and made verifiable. The present work focuses on this scenario, considering the construction of article-level knowledge structures and the role of human feedback in supporting reliable interpretation of scientific content.

3 Preliminaries

Scientific publications contain structured argumentative information that can be formalized for computational extraction and reasoning. This section introduces the formal notation used throughout the paper.

3.1 Scientific Text and Document Structure

Let a scientific document be denoted as

$$D = \{s_1, s_2, \ldots, s_n\},$$

where s_i represents the i-th sentence. Each sentence belongs to a rhetorical section

$$\rho(s_i) \in \{\text{Abstract, Introduction, Methods, Results, Discussion, Conclusion}\}.$$

Scientific discourse typically exhibits an empirical structure linking hypotheses, experiments, and conclusions. Following prior literature in scientific argument mining, a sentence s_i may serve one

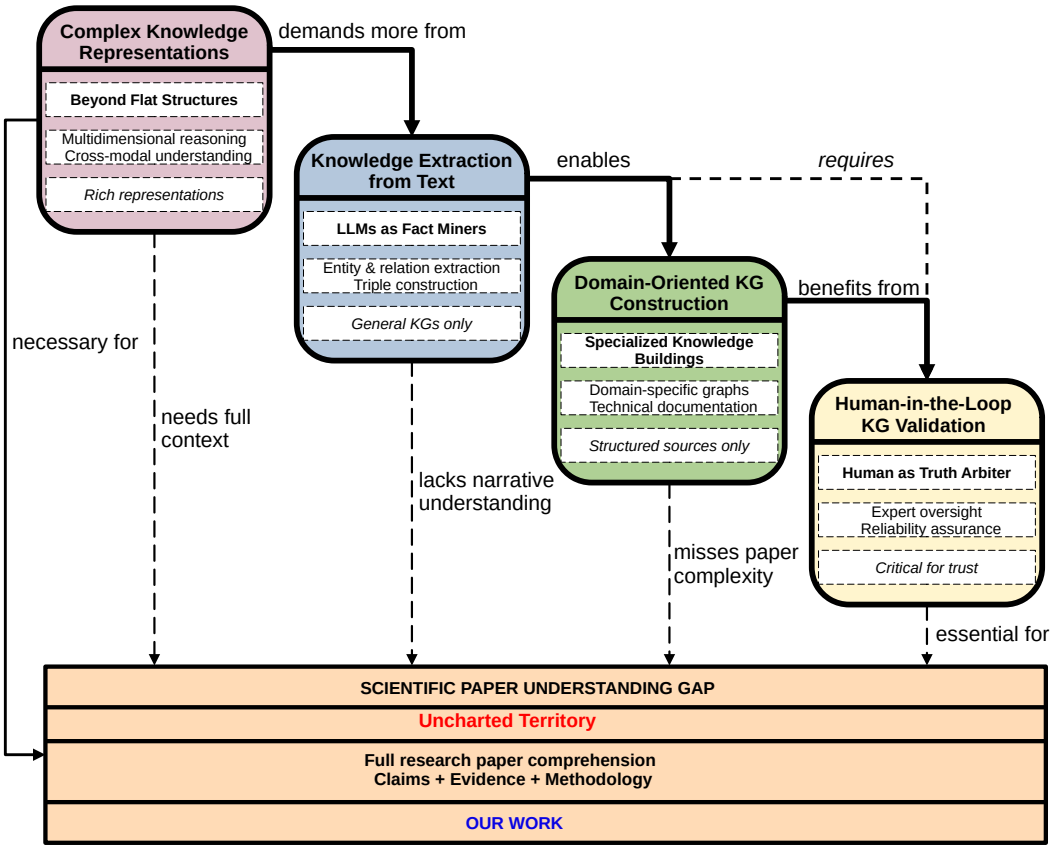


Figure 1: Conceptual map of the current research landscape at the intersection of LLMs and KGs.

or more rhetorical roles:

$$\tau(s_i) \in \{\text{Claim,Evidence,Method,}\atop\text{Assumption,Limitation,Background}\}.$$

3.2 Scientific Claims

A scientific claim is formalized as a logical proposition grounded in experimental or theoretical reasoning [14]. Let C denote the set of claims identified in D. Each claim $c \in C$ is a tuple

$$c = \langle \phi, \gamma, \sigma \rangle,$$

where

- ϕ is a natural language statement expressing a research conclusion or hypothesis,
- $\gamma \subseteq D$ is a set of supporting spans,
- $\sigma \in [0, 1]$ is a system-estimated confidence score.

Claims may involve comparative or causal relations. A binary scientific claim can be represented as:

$$\phi : x \xrightarrow{r} y,$$

where x and y denote scientific entities (e.g., models, variables, datasets) and r denotes a scientific relation (e.g., "outperforms", "correlates with", "reduces", "causes").

3.3 Evidence and Provenance Grounding

Evidence consists of document spans supporting a claim. Let E be the set of evidence units. Each unit $e \in E$ is defined as:

$$e = \langle t, \rho(t) \rangle, \quad t \in D,$$

where t is a contiguous span of one or more sentences.

A claim–evidence alignment function

$$\mathcal{A} : C \to 2^E$$

assigns to each claim a subset of evidence text segments. Evidence must originate from the source document:

$$\forall e \in \mathcal{A}(c),\ e \subseteq D.$$

3.4 Scientific Knowledge Graph

The extracted scientific knowledge is represented as a heterogeneous graph,

$$G = (V, E_R),$$

where

$$V = C \cup E \cup M,$$

and M denotes contextual nodes (e.g., methods, datasets, metrics, tasks).

Edges capture semantic relations:

$$E_R = \{(u, r, v) \mid u, v \in V,\ r \in \mathcal{R}\},$$

with $\mathcal{R}$ including:

$\mathcal{R}$ = {supports, refutes, uses, reports, evaluated_on, measured_by}.

A core relation is the evidence-support link:

$$(c, \text{supported_by}, e) \in E_R.$$

This formulation extends conventional triple-based KGs by incorporating explicit provenance and scientific discourse structure.

3.5 LLM-Driven Extraction Model

Let f_θ denote a large language model parameterized by θ. Given a document D, the model predicts claims and evidence spans:

$$f_\theta(D) \to (C, E, \mathcal{A}).$$

Retrieval grounding is incorporated through a scoring function

$$\text{score}(t, c) = \text{sim}(\text{emb}(t), \text{emb}(c)),$$

where emb$(\cdot)$ denotes contextual embeddings, and sim denotes cosine similarity or a learned relevance function. Human feedback is applied selectively to low-confidence pairs:

$$\mathcal{H} = \{(c, e) : \sigma(c) < \tau_\sigma \vee \text{score}(e, c) < \tau_s\}.$$

This setup enables selective expert verification while preserving autonomy for high-confidence extractions.

4 Methodology

The extraction pipeline is organized into four logically consecutive stages: document ingestion, LLM-driven semantic processing, JSON-based structured representation, and graph-oriented postprocessing. These stages collectively produce a deterministic transformation

$$D_{\text{PDF}} \longrightarrow T_{\text{clean}} \longrightarrow J_{\text{extract}} \longrightarrow G_{\text{KG}},$$

where D_{PDF} denotes the raw scientific article, T_{clean} is the normalized text, J_{extract} represents the structured JSON output, and G_{KG} is the final graph representation. The choices at each stage are shaped by the requirements of scientific information extraction and the need for reliable, provenance-aware knowledge graph construction.

Document Loading and Normalization

The uploaded scientific article in PDF format is first converted into a clean, machine-readable sequence of sentences:

$$T_{\text{clean}} = \{s_1, s_2, \ldots, s_n\}.$$

The extraction process applies layout-aware text parsing, removes structural noise such as headers and footers, repairs hyphenated line breaks, and performs sentence segmentation. Because scientific PDFs frequently exceed the context limits of large language models, the system partitions the text into overlapping windows,

$$C_i = \{s_k, s_{k+1}, \ldots, s_{k+m}\},$$

with a stride $\delta < m$ to preserve local semantic coherence. Formally, the document is covered by windows satisfying

$$\bigcup_i C_i = T_{\text{clean}}, \qquad C_i \cap C_{i+1} \neq \varnothing,$$

ensuring that entities near window boundaries remain recoverable.

GPT-Based Semantic Processing

After chunking, each segment C_i is processed by GPT-5 using a constrained extraction prompt designed to enforce grounded, non-hallucinatory, and schema-aligned behavior. The transformation from unstructured scientific PDFs to schema-aligned, evidence-constrained JSON extractions is illustrated in Figure 2.

Unlike generative prompting strategies, the objective here is not to summarize or reinterpret the content but to perform a deterministic semantic mapping

$$f_{\text{LLM}} : C_i \mapsto J_i,$$

where the output J_i is a JSON object conforming to a predefined schema of claims, evidence fragments, methodological descriptors, datasets, metrics, and numerical values. The prompt explicitly restricts the model to extracting information *only* from text present in C_i and prohibits the introduction of external knowledge, rephrased interpretations not grounded in text, or implicit inferencing beyond what is explicitly written.

A scientific claim is represented as a structured tuple

$$q = (t, e, M, \Delta, \Theta, R),$$

where:

- t is a controlled paraphrase of the claim that preserves its logical content,
- e is a set of verbatim evidence snippets quoting the source text,
- M is a set of methodological descriptors (e.g., model variants, training regimes, prompting modes),
- Δ is the set of datasets explicitly mentioned,
- Θ is the set of evaluation metrics reported,
- R is a set of numerical results of the form

  $$R = \{(m, v, c)\}, \quad m \in \Theta,\ v \in \mathbb{R} \cup \mathbb{R}^2,\ c \in \text{Context},$$

  where v may represent either a scalar value or a range, and c denotes the contextual clause (e.g., dataset split, model variant, training configuration).

The JSON schema enforces that each extracted claim be "atomic"—that is, each q must represent a single, indivisible scientific statement rather than a compound sentence containing multiple findings. Atomicity is operationalized by requiring the model to emit multiple claim objects if a single sentence contains multiple logical propositions.

A central constraint of the prompt is that for every extracted field x in J_i, the corresponding evidence must be recoverable as a verbatim substring of C_i:

$$\forall x \in J_i, \qquad x.\text{evidence} \subseteq C_i.$$

This constraint is articulated explicitly in the prompt through rules that penalize hallucination and enforce provenance. The model is required to return one or more exact textual spans supporting each element of the tuple q, including numeric values and methodological attributes. This constraint yields three benefits: (i) extraction becomes auditable, (ii) downstream fact-checking is simplified, and (iii) subsequent human-in-the-loop validation can verify the correctness of each element through direct alignment with the source text.

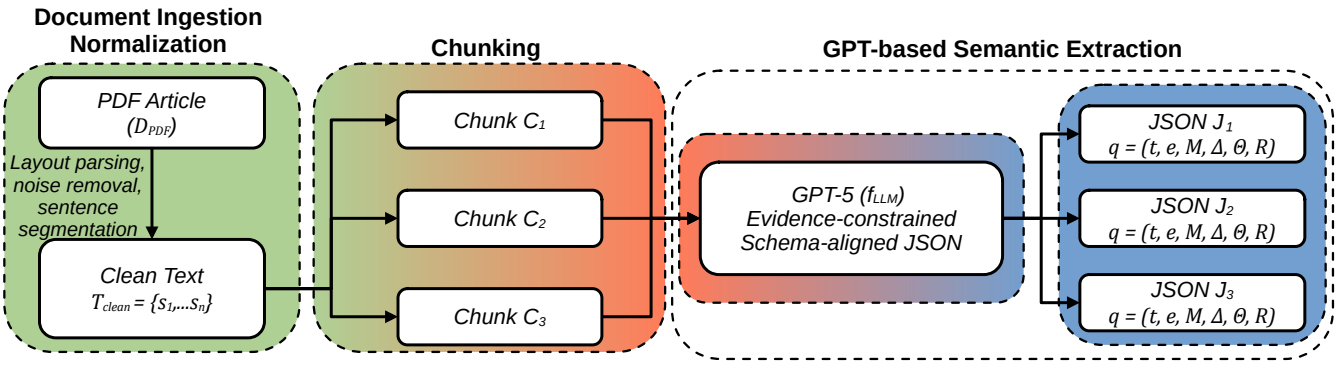


Figure 2: End-to-end extraction pipeline from raw scientific PDFs to structured, evidence-grounded JSON representations.

To ensure structural consistency across all outputs, the prompt includes a JSON template, type constraints, and cardinality restrictions. Formally, the model must satisfy:

$$\text{schema}(J_i) = \text{schema}_{\text{target}},$$

where $\text{schema}_{\text{target}}$ defines field presence, allowed entity types, and permissible formats for numerical values. GPT-5 is also instructed to avoid coercing missing values into default placeholders; instead, fields must be omitted or returned as empty sets when not present in C_i.

The mapping f_{LLM} requires the model to perform lexically faithful extraction. To formalize this, let $\text{span}(C_i)$ denote the set of all textual substrings of C_i. Then GPT-5 must ensure that:

$$e \subseteq \text{span}(C_i), \qquad M \subseteq \text{span}(C_i),$$
$$\Delta \subseteq \text{span}(C_i), \qquad \Theta \subseteq \text{span}(C_i).$$

Only the paraphrased claim text t is allowed to be model-generated, and even there the semantic divergence must satisfy:

$$\text{sim}(t, t_{\text{orig}}) > \tau,$$

for a high similarity threshold $\tau \approx 0.90$, computed using sentence embeddings during validation.

To prevent the model from producing unstated implications, the prompt includes a prohibition of inferential reasoning:

$$\nexists\, x \in J_i : x \text{ is derivable only through external knowledge.}$$

This rule ensures that only explicitly stated numerical results, methods, or evaluations are included. For example, the model is not allowed to infer that a dataset is "medical" or that a model is "transformer-based" unless the text states so explicitly.

The per-chunk results J_i are aggregated into a document-level JSON $J = \bigcup_i J_i$ without modification. Since each claim includes its own evidence spans, downstream components can trace each extraction back to its source segment C_i, enabling chunk-level provenance and cross-validation.

This controlled extraction design ensures that GPT-5 operates as a deterministic information extractor rather than a generative agent, producing structured, verifiable, and provenance-aligned scientific statements suitable for later normalization and graph construction.

JSON Creation and Evidence-Grounded Structuring

The JSON format was selected to provide a flexible but stable schema across chunks and to simplify aggregation. Each chunk produces

$$J_i = \{q_1, q_2, \ldots, q_{k_i}\},$$

where all q_j share the same structure, enabling machine-level consistency. After extraction, all JSON objects from the document are merged by canonicalizing equivalent entities. For two extracted entities e_p and e_q, the system computes a similarity score,

$$\text{sim}(e_p, e_q) = 1 - \frac{\text{Lev}(e_p, e_q)}{\max(|e_p|, |e_q|)},$$

where Lev denotes the Levenshtein distance. Entities satisfy a canonicalization condition

$$\text{sim}(e_p, e_q) > \tau \quad \Rightarrow \quad e_p \equiv e_q,$$

while ambiguous cases in an interval $[\tau_1, \tau_2]$ are resolved using GPT-5 as a semantic disambiguator. The resulting canonical entity set E becomes the foundation for graph construction.

The JSON aggregation and NLP processing workflow—including entity canonicalization, normalization and typing, relation extraction, and claim deduplication—is illustrated in Figure 3

To illustrate the structured representation used throughout the extraction pipeline, a minimal JSON example is provided in Listing 1.

Listing 1: Example of the JSON structure used for extracted scientific entities from [7]

```
"claims": [
  {
    "text": "Instruct models without fine-tuning
         perform poorly on strict triple-level
        KGC; LoRA adaptation is essential to
        reach strong performance.",
    "evidence": "  The   results for the
        instruction models that have not
        undergone fine-tuning are significantly
        lower, compared to their fine-tuned
        counterparts.   ",
    "methods": [
```

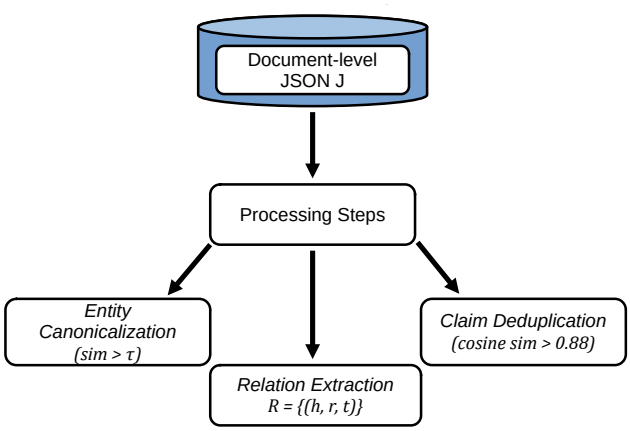


Figure 3: JSON aggregation and NLP processing pipeline applied to chunk-level outputs.

```
        "Zero/few-shot prompting of instruction-tuned LLMs (no parameter updates)",
        "Direct comparison to LoRA-adapted versions under identical prompts",
        "Strict micro-F1 computation on exact triples"
      ],
      "datasets": [
        "ADE",
        "CoNLL04",
        "SciERC"
      ],
      "metrics": [
        "micro-F1"
      ],
      "numerical_results": [
            {
        "metric": "micro-F1",
        "value": 0.333,
        "condition": "Mistral-7B-Instruct-v0.3, ADE, best base setting"
        },
        ...
        {
        "metric": "micro-F1",
        "value": 0.592,
        "condition": "Llama-3.1-8B-Instruct, CoNLL04, best fine-tuned natural-language setting"
        }
  ]
```

4.1 NLP Processing of Extracted JSON Structures

After LLM extraction yields a JSON of claims, evidence, methods, datasets, metrics, and values, an NLP phase normalizes entities, resolves coreference, and extracts latent relations—operating solely on the JSON without requerying the source. Though human-readable, the raw fields exhibit linguistic variability and unaligned surface forms, which are standardized via automated NLP steps.

All surface mentions of models, datasets, metrics, and methodological components are first standardized. Token-level normalization performs Unicode normalization, lowercasing, scientific-domain lemmatization, removal of determiners, and multi-word expression collapsing. Given a raw token sequence x, the normalization operator $\mathcal{N}(x)$ is defined as:

$$\mathcal{N}(x) = \mathrm{MWU}(\mathrm{lemma}(\mathrm{clean}(x))),$$

where MWU converts multi-word scientific expressions (e.g., "Mistral-7B-Instruct-v0.3 fine-tuned with LoRA") into canonical single-unit forms.

Each normalized phrase is mapped to one of the canonical entity types:

$$T \in \{\mathrm{Model},\ \mathrm{Dataset},\ \mathrm{Metric},\ \mathrm{Method},\ \mathrm{Claim}\}.$$

Classification uses a hybrid approach combining rule-based identification (e.g., dataset name patterns), transformer embedding similarity for model families, metric-recognition regular expressions, and heuristic fallbacks for methodological phrases.

Claim sentences frequently contain implicit relations such as performance comparisons, causal assertions, and model–dataset associations. To convert these into explicit graph edges, the pipeline applies a dependency parser with semantic role labeling (SRL). Given a claim sentence s, the relation extraction operator $\mathcal{E}(s)$ yields a set of directed tuples:

$$R = \mathcal{E}(s) = \{(h, r, t)\},$$

where h is the head entity (typically a model), r the relation label (*outperforms*, *evaluated-on*, *improves*), and t the target entity (dataset, metric, or numeric value).

For example, the statement "Fine-tuned lightweight LLMs outperform CodeKGC by 15–20 F1 points on ADE" produces:

$$(\mathrm{LLM}_{\mathrm{light}},\ \texttt{outperforms},\ \mathrm{CodeKGC}),$$

and a metric relation:

$$(\mathrm{LLM}_{\mathrm{light}},\ \texttt{improvement},\ \mathrm{F1{:}\ 15{-}20}).$$

Numerical spans in `numerical_results` and in free-text claims are normalized into structured objects. The system identifies absolute metric values, deltas, and ranges represented as $(v_{\min}, v_{\max})$. A normalized tuple takes the form:

$$\langle m, v, c \rangle,$$

where m is the metric, v the extracted numeric value, and c the associated experimental condition.

Graph Construction

After aggregation, the system constructs an intermediate protograph,

$$G = (V, E),$$

where V is the set of canonicalized entities and E the set of relations supported by evidence. For a scientific claim q, the system generates relational triples by interpreting interactions among its fields (e.g., method–dataset, method–metric, model–result). Each relation is represented as

$$(h, r, t, e_{\mathrm{src}}),$$

where h is the head entity, t the tail entity, r the relation type, and e_{src} the supporting quotation. A triple is accepted into the graph if and only if its evidence explicitly supports it:

$$(h, r, t) \in E \quad \Longleftrightarrow \quad e_{\text{src}} \vDash (h, r, t),$$

where $\vDash$ denotes textual entailment. For cases where support is uncertain, GPT-5 performs a binary "supported / unsupported" verification.

The final knowledge graph G_{KG} is exported in machine-readable formats such as GraphML, JSON-LD, Neo4j import sets, or CSV triples, enabling downstream querying, reasoning, and visualization.

5 Experiments and Results

In this section aims to empirically assess the proposed extraction pipeline from PDF articles to a provenance-aware scientific knowledge graph. The experiments are designed to answer the following research questions:

- **RQ1 (Extraction quality).** How accurately does the pipeline recover entities, claims, and relations compared to extraction baselines?
- **RQ2 (Evidence grounding).** To what extent are extracted structures reliably supported by verbatim evidence spans in the source document?
- **RQ3 (Graph quality).** Does the constructed knowledge graph preserve the structure of the underlying scientific discourse, and how does it compare to graphs produced by baseline systems?

5.1 RQ1: Extraction Quality

To assess how reliably the system recovers factual scientific assertions from real articles, a subset of ten papers from the intersection of large language models and knowledge graphs was manually annotated [5, 7, 16, 18, 24, 28, 29, 32, 38, 46]. Each document was labelled at the fact level: annotators identified atomic scientific claims, associated models, datasets, metrics, methodological configurations, and numerical results when explicitly stated in the text.

Since claims are natural language statements, evaluation could not rely on span-level matching or surface-form overlap. Instead, for every predicted claim sentence t_{pred} a semantic similarity score with each gold claim t_{gold} was computed using sentence embeddings. A claim was considered correct only if one gold claim satisfied

$$\text{sim}(t_{\text{pred}}, t_{\text{gold}}) > 0.88,$$

which captures agreement in meaning even when phrasings differ. For claims that included structured fields—such as model, dataset, metric, or numeric value—the match was counted only if the textual similarity condition was met *and* the corresponding entities aligned with the gold annotation after canonicalization. This ensured that a claim was not merely paraphrased correctly, but also grounded in the same scientific components used in the article.

To construct a reliable evaluation set, each of the ten papers was independently reviewed by two annotators with background in information extraction. The annotators were instructed to identify only atomic claims that satisfy two conditions: (i) the statement must express a verifiable scientific finding involving at least one model, dataset, metric or methodological setup; and (ii) the statement must be explicitly present in the text of the PDF, such that its supporting span can be quoted verbatim.Some examples

Claims that relied on inference, prior knowledge, or implicit interpretation were not included. Whenever multiple propositions appeared within one sentence, they were split into separate atomic claims to align with the format of the JSON outputs produced by the extraction pipeline. All candidate claims were then consolidated via sentence-level embedding clustering to remove near-duplicates, after which disagreements between annotators were resolved through adjudication.[1]

Using this criterion, 143 gold claims and 367 fact-level relations were evaluated. Claim-level F1 reached 73.7% when matching only the textual assertion and 59.9% when requiring alignment of both meaning and structured components. To demonstrate how the same evidence can yield multiple interpretations, one annotated sentence [7] was selected from the corpus:

> *"Training lightweight LLMs to generate chain-of-thought rationale before triples generally degrades KGC performance after fine-tuning."*

Table 1 shows how GPT produced different claim candidates based on this evidence. Only one matched the gold annotation.

Table 1: Different GPT interpretations of the same evidence span.

Type	Generated Claim	Similarity	Matched
Correct	*"Chain-of-thought prompting harms triple-level KGC accuracy when applied to small LLMs."*	0.91	Yes
Partially correct	*"Chain-of-thought improves interpretability but does not always improve KGC performance."*	0.84	No
Incorrect (inverted)	*"Generating explanations before triples boosts KGC performance on lightweight models."*	0.59	No
Overly general	*"Reasoning strategies often affect KGC quality in unpredictable ways."*	0.51	No

Most failed matches fall into two categories: either no explicit metric/value was present (making the claim incomplete under the matching criterion), or the predicted sentence expressed a general observation rather than an experimentally supported finding. Claims reported only in tables were also occasionally missing when numerical references were abbreviated or split across latex formatting in the PDF.

5.2 RQ2: Evidence Grounding

The assessment of evidence grounding focused on whether extracted claims and entities were supported by verbatim spans in the original PDF. Each predicted field was therefore required to include an evidence string, and evaluation measured both evidence

[1] Inter-annotator agreement before adjudication was 0.82 Cohen's κ.

precision (correctly localized spans over all predicted spans) and evidence coverage (gold annotations that were successfully recovered with at least one valid span). This evaluation was conducted on the same ten annotated articles used for RQ1.

Evidence precision reached 81.3%, while coverage amounted to 74.6%, indicating that most predictions were not only correct but also properly grounded in the document. Errors were predominantly due to partial span boundaries or inclusion of adjacent phrases; fully incorrect evidence was rare. Table 2 summarizes the most common categories of mismatches.

Table 2: Frequent evidence-grounding errors across the corpus.

Error Type	Example Extracted Evidence	Share (%)
Boundary mismatch	*"...their fine-tuned counterparts also show improved..."* (gold span excluded last clause)	41.2
Overly broad span	*"Results for all instruction models without fine-tuning are shown to be lower than base versions trained with LoRA"*	33.5
Wrong sentence	Mentioned dataset name but no experimental result	17.8
Unsupported (hallucinated)	Mentioned metric not present in the paper	7.5

One successful recovery is illustrated below from [16]:

> *Claim: "The experimental dataset consists of 480 multimodal equipment fault tickets and five Excel sheets with component-level numerical attributes for a bridge crane, from which 4,585 triples and multimodal hypernodes are constructed."*
> **Matched evidence:** *"Our dataset contains 480 multimodal equipment fault tickets and five XLS files with detailed numeric attributes for each bridge crane component, resulting in a total of 4,585 triples and multimodal hypernodes."*

Here the model correctly identified not only the quantitative components but also the structural implication ("multimodal hypernodes").

In contrast, an incorrect grounding case is shown below from [32]. Although the structured claim was valid, the evidence spans combined content from two separate parts of the article that were not adjacent in the source text:

> *Claim: "Computer Science Knowledge Graph (CS-KG) is a large scientific KG generated from 6.7M publications, containing about 41M statements represented as 350M triples and about 10M entities typed into five categories and 179 relations."*
> **Extracted evidence (invalid):** *"CS-KG… comprises 41 million statements represented as 350 million triples within the Computer Science domain." and "The knowledge graph contains about 10M entities, classified within the five categories Method, Task, Material, Metric, and OtherEntity, and connected through 179 object properties."*

This example demonstrates a typical failure mode: GPT concatenated two non-contiguous spans to construct the evidence, even though both sentences exist separately in the PDF.

5.3 RQ3: Graph Quality

The third question evaluates the quality of the final document-level knowledge graphs produced from the aggregated JSON structures. Quality was measured across three dimensions: (i) strict triple-level F1 for graph edges; (ii) claim coverage — the proportion of annotated atomic claims that appeared as at least one triple in the graph; (iii) query accuracy — the ability to answer templated questions such as "Which model achieves the highest F1 on dataset d?" or "Which prompting strategy was compared to LoRA on SciERC?".

The resulting graphs (an example shown in Figure 4) preserved the major structural components of the scientific discourse in article, in this case by Shi et al. [28]. Model–dataset–metric subgraphs emerged naturally, revealing experimental comparisons across methods and configurations. Evidence-aware extraction produces graphs that not only capture structural relationships, but also retain sufficient contextual information to support meaningful downstream analysis of LLM–KG research.

6 Discussion

6.1 Limitations

Evidence fragmentation emerges as main constraint. Despite overlapping window segmentation preserving local coherence ($C_i \cap C_{i+1} \neq \varnothing$), 33.5% of evidence spans were overly broad and 7.5% contained hallucinated content (Table 2). The pipeline fails to handle non-contiguous evidence [20] effectively, as demonstrated when GPT concatenated two separate sentences to construct invalid evidence for the CS-KG claim. This violates the provenance principle requiring $\forall e \in \mathcal{A}(c), e \subseteq D$. PDF formatting complexities [22] further exacerbate this issue, with table-based results and split LaTeX expressions causing systematic failures in numerical extraction, particularly when value-condition alignment is required.

Structural rigidity in the atomic claim representation ($q = (t, e, M, \Delta, \Theta, R)$) limits expressiveness. The requirement for full structured alignment achieved only 59.9% F1, lower than textual similarity alone (73.7%). This gap reflects how scientific discourse frequently embeds multiple propositions within single sentences (e.g., "LoRA fine-tuning improves performance *but* increases inference latency by 15%"). The prohibition of implicit inference ($\nexists x \in J_i$: x is derivable only through external knowledge) excludes valid scientific implications, as evidenced by GPT generating both correct and inverted interpretations from identical evidence spans (Table 1).

Domain dependence challenges generalizability. Evaluation on ten LLM-KG papers showed strong evidence precision (81.3%), but the pipeline's reliance on GPT-5 creates three bottlenecks: (1) proprietary model dependency limits reproducibility [33], (2) context window constraints necessitate aggressive chunking [25] that disrupts discourse flow across rhetorical sections ($\rho(s_i)$), and (3) evidence constraint templates require re-engineering for domains with different argumentative structures [43]. Entity canonicalization further compounds this issue, as Levenshtein-based matching

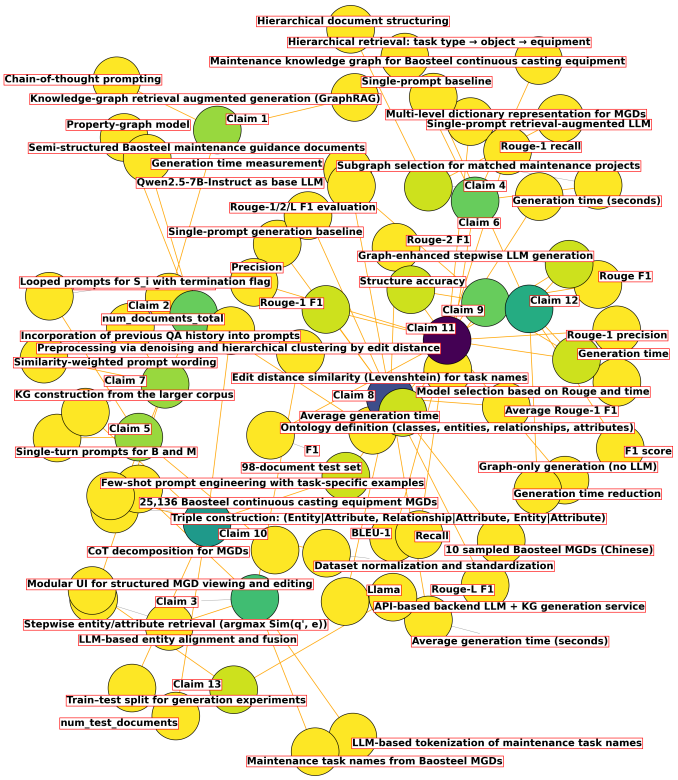


Figure 4: Graph for the scientific article by Shi et al. [28].

$(\text{sim}(e_p, e_q) > \tau)$ fails for coreferential entities with divergent surface forms (e.g., "BERT" versus "Bidirectional Encoder Representations from Transformers").

6.2 Future Work

Adaptive document understanding should replace fixed windowing with rhetorically-aware segmentation [19]. Instead of uniform chunks ($C_i = \{s_k, \ldots, s_{k+m}\}$), future systems could leverage section headers ($\rho(s_i)$) and discourse markers to create variable-length segments preserving logical units spanning multiple sections. Complementing this, *table-specific extraction modules* using vision-language models could recover structured results currently missed during PDF normalization.

Hybrid extraction architectures offer a path beyond GPT dependency. Recent work on lightweight LLMs [7] suggests fine-tuning open-weight models (e.g., Mistral-7B) with explicit evidence grounding objectives could achieve comparable precision at lower computational cost. Crucially, such systems should incorporate *verifiable inference*: rather than prohibiting all implicit reasoning, they could flag derived claims with explicit provenance trails like $(c, \text{derived_from}, \{e_1, e_2\})$, addressing the 41.2% boundary mismatch errors while maintaining auditability.

Dynamic schema evolution would enhance cross-domain applicability. Instead of static schemas ($\text{schema}(J_i) = \text{schema}_{\text{target}}$), future pipelines should employ meta-learning to adapt claim structures to domain conventions [1, 10]. For instance, biomedical papers often embed causal claims ("drug X *induces* protein Y expression") requiring different relation types than comparative computer science claims. Semi-supervised relation extraction guided by domain ontologies could automatically expand $\mathcal{R} = \{\text{supports}, \text{uses}, \ldots\}$ while preserving grounding constraints.

Targeted human verification [8, 12] presents the major improvement opportunity. Current human feedback thresholds ($\mathcal{H} = \{(c, e) : \sigma(c) < \tau_\sigma\}$) trigger verification only for low-confidence extractions. Future systems should prioritize *high-impact verification*: automatically flagging claims with high centrality in G_{KG} or those containing critical numerical results ($R = \{(m, v, c)\}$). This approach could reduce hallucination rates [13, 26] while optimizing expert time allocation.

6.3 Concluding Perspective

The tension between *extractive fidelity* [41] and *scientific expressiveness* [31] represents a fundamental challenge in automated knowledge graph construction. Future progress requires reconciling strict provenance requirements with the nuanced reasoning patterns inherent in scientific discourse. Rather than relaxing grounding constraints, the path forward lies in developing representations that accommodate domain-specific argumentation while maintaining verifiable connections to source text [35]. As autonomous scientific knowledge bases evolve, such systems must ultimately balance mechanical reliability with the interpretive depth that characterizes scientific advancement. The framework presented here establishes a foundation for evidence-grounded extraction, but realizing its full potential necessitates addressing these limitations through adaptive architectures and domain-aware modeling.

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